Given that $pK_a = 10.26$

$pK_a = -\log K_a$

i.e., K_a = antilog of $(-pK_a)$

= antilog of (-10.26)

= antilog of (-11 + 0.74)

$= 10^{-11} \times 5.5$

[antilog of 0.74 = 5.49 $\simeq$ 5.5]

ii) $K_h = \frac{K_w}{K_a} = \frac{1 \times 10^{-14}}{5.5 \times 10^{-11}}$

$= 1.8 \times 10^{-4}$

iii) $pH = 7 + \frac{pK_a}{2} + \frac{\log C}{2}$

$= 7 + \frac{10.26}{2} + \frac{\log 0.05}{2} = 7 + 5.13 - 0.65$

$= 11.48$

Unit 9 Electro Chemistry

1. 1F = 96500 C = 1 mole of e^- = 6.023×10^{23} e^-

$\therefore$ 9650 C = $\frac{6.22 \times 10^{23}}{96500} \times 9650 = 6.022 \times 10^{22}$

Option (C)

2. $Mn^{2+} + 2e^- \rightarrow Mn\ (E^\circ_{red}) = -1.18V$

$2\left[Mn^{2+} \rightarrow Mn^{3+} + e^-\right]\left(E^\circ_{ox}\right) = -1.51V$

$3Mn^{2+} \rightarrow Mn + 2Mn^{3+}$ $E^\circ_{cell} = ?$

$E^\circ_{cell} = \left(E^\circ_{ox}\right) + \left(E^\circ_{red}\right)$

$= -1.51 - 1.18$ and non spontaneous

$= -2.69V$

Since E° is –ve ΔG is +ve and the given

forward cell reaction is non – spontaneous.

(Option (b))

3. Anodic oxidation: (Reverse the given reaction)

$\left(E^\circ_{ox}\right) = 0.76V$ cathodic reduction

$\therefore$ $E^\circ_{cell} = \left(E^\circ_{ox}\right) + \left(E^\circ_{red}\right)$

= 0.76 + 0.34 = 1.1V

(Option (c))

4. $\Lambda = \frac{\kappa}{M} \times 10^{-3}\ mol^{-1}\ m^3$

$= \frac{5.76 \times 10^{-3}\ S\ c\ m^{-1} \times 10^{-3}}{0.5} mol^{-1}\ m^3$

$= \frac{5.76 \times 10^{-3} \times 10^{-3} \times 10^6}{0.5}\ S\ cm^{-1} mol^{-1}\ cm^3.$

$= 11.52\ S\ cm^2\ mol^{-1}$

(Option (b))

5. $(\Lambda_\infty)_{HoAC} = \left[(\Lambda^\circ)_{HCl} + (\Lambda^\circ)_{NaOAC}\right] - (\Lambda^\circ)_{NaCl}$

$= (426.2 + 91) - (126.5)$

$= 390.7$

(Option (c))

6. 1F = 96500 C = charge of 1 mole of e^- = $charge\ of\ 6.022 \times 10^{23}\ e^-$

(Option (b))

7. $7MnO_4^- + 5e^- \rightarrow Mn^{2+} + 4H_2O$

5 moles of electrons i.e., 5F charge is required.

(Option (a))

8. m=ZIt

$= \frac{40 \times 3.86 \times 2500}{2 \times 96500}$

= 2g

41min 40sec = 2500 seconds

$Z = \frac{m}{n \times 96500} = \frac{40}{2 \times 96500}$

(Option (b))

9. m=ZIt (mass of 1 mole of Cl_2 gas = 71)

$t = \frac{m}{ZI}$ ($\therefore$ mass of 0.1mole of Cl_2 gas = 7.1 g mol^{-1})

$= \frac{7.1}{\frac{71}{2 \times 96500} \times 3} (2\ Cl^- \rightarrow Cl_2 + 2e^-)$

$= \frac{2 \times 96500 \times 7.1}{71 \times 3}$

$= 6433.33 sec$

$= 107.2\ min$

(Option (b))

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